

Total No. of Questions : 8]

SEAT No. :

PD4708

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[6404]-214

B.E. (Mechanical)

OPERATIONS RESEARCH

(2019 Pattern) (Semester - VII) (Elective - IV) (402045D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1. or Q2, Q3. or Q4, Q5. or Q6, Q7. or Q8.
- 2) Answers in One answer Books.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary

Q1) a) A manufacturer of furniture makes two products - chairs and tables. Processing of this product is done on two machines A and B. A chair requires 2 hours on machine A and 6 hours on machine B. A table requires 5 hours on machine A and no time on machine B. There are 16 hours of time per day available on machine A and 30 hours on machine B. Profit gained by the manufacturer from a chair and a table is Rs 2 and Rs 10 respectively. What should be the daily production of each of two products? [12]

b) Explain with the help of an example general syntax of LPP. [6]

OR

Q2) a) Use two-phase simplex method to solve [12]

Maximize $Z = 5X_1 + 3X_2$

Subject to Constraints

$$2X_1 + X_2 \leq 1;$$

$$X_1 + 4X_2 \geq 6;$$

$$X_1, X_2 \geq 0;$$

b) Discuss the following [6]

i) Objective function

ii) Constraints

iii) Feasible and infeasible solution set

P.T.O.

- Q3) a)** Consider a problem of assigning four clerks to five tasks. The time (hrs) required to complete the task is given below. Clerks II cannot be assigned to task C and Clerk 3 cannot be assigned to task A. Find the optimum schedule. **[10]**

		Tasks				
		A	B	C	D	E
Clerks	I	9	11	15	10	11
	II	12	9	----	10	9
	III	----	11	14	11	7
	IV	14	8	12	7	8

- b)** Find out the initial feasible solution for the transportation problem given below. Using **[8]**

- North- west corner rule
- VAM method
- Least cost method

		Destinations			
		1	2	3	4
Plants	1	2	3	11	7
	2	1	0	6	1
	3	5	8	15	9
Requirement		7	5	3	2
		Supply			
		6	1	10	

OR

- Q4) a)** What do you mean by balancing an assignment problem? What steps you take to solve maximization case in assignment problem? Explain. [6]
- b)** A manager has 4 jobs on hand to be assigned to 3 of his clerical staff. Clerical staff differs in efficiency. The efficiency is a measure of time taken by them to do various jobs. manager wants to assign the duty to his staff, so that the total time taken by the staff should be minimum. The matrix given below shows the time taken by each person to do a particular job. Help the manager in assigning he jobs to the personnel. [12]

Jobs.	Men (time taken to do job in hours)		
	X	Y	Z
A	10	27	16
B	14	28	7
C	36	21	16
D	19	31	21

- Q5) a)** The following mortality rates have been observed for a certain type of fuse. [12]

Week	1	2	3	4	5
% failing by the End of Week	5	15	35	75	100

There are 1000 fuses in use and it costs Rs 5 to replace an individual fuse. If all fuses were replaced simultaneously it would cost Rs 1.25 per fuse. It is proposed to replace all fuses at fixed intervals of time, whether or not they have burnt out and to continue replacing burnt out fuses as they fail. At what intervals the group replacement should be made? Also prove that this optimal policy is superior to the straight forward policy of replacing each fuse only when it fails.

- b)** Differentiate between CPM and PERT analysis of project network. [5]

OR

Q6) a) A project has the following data.

[12]

Task	1-2	1-3	1-4	2-5	2-6	3-6	4-7	5-7	6-7
Optimistic Time	5	18	26	16	15	6	7	7	3
Pessimistic time	10	22	40	20	25	12	12	9	5
Most likely time	8	20	33	18	20	9	10	8	4

Determine the following:

- Construct the network and find out the \Expected task time and their variance.
- The earliest and the latest expected completion times of each event.
- The critical path.
- The probability of an event occurring at the expected completion date if the original scheduled time of completing the project is 41.5 weeks.
- The duration of the project that will have 95 % chance of being completed. Use following table

Probability	0.3085	0.3050	0.3015	0.2981	0.95
Normal distribution constant (Z)	-0.50	-0.51	-0.52	-0.53	1.64

- Explain what you mean by crashing of the network and various flaws in the network construction.

[5]

Q7) a) Explain various steps involved in the simulation process. **[5]**

- b) Due to the new technology up graduation done on the plant of the auto component manufacturing firm. It has come up with a need of training its workforce in the different trainings. It thus contracted about four different training firms for providing the training in technical domain. The results for all the trainings was compiled for knowledge enhancement on a scale of 100 as follows: **[12]**

Number of Courses	Firm 1	Firm 2	Firm 3	Firm 4
0	0	0	0	0
1	25	20	40	10
2	50	70	60	20
3	60	90	80	30
4	80	100	100	40
5	100	100	100	50
6	100	100	100	60
≥ 7	100	100	100	70

How many courses should be allocated to each firm in order to maximize knowledge?

OR

Q8) a) Explain the characteristics of dynamic programming in brief. **[5]**

b) Students arrive at the head office of www.gunjan.com. According to the following arrival and service time probability distributions are as follows: **[12]**

Inter-arrival time (min)	Frequency	Service time (min)	Frequency
1	1	1	1
2	4	2	4
3	7	3	7
4	17	4	17
5	31	5	31
6	23	6	23
7	7	7	7
8	5	8	5
9	3	9	3
10	2	10	2

Use Monto Carlo Simulation and determine the following

- The average waiting time before service.
- The average time a student spends in a system.

Use following random numbers : 17,86,84,79,33,55,6,42,93,38

